CubeSat Electrical Power System (EPS) – Project Report

Project Overview

Our senior design project focuses on the development of a CubeSat Electrical Power System (EPS) — a miniature version of the system that manages power generation, storage, and distribution in real satellites. The main goal is to design and implement a functional prototype that demonstrates how solar energy can be harvested, stored in rechargeable batteries, and supplied to different onboard subsystems efficiently. This mock-up will serve as a realistic educational model of a CubeSat's power subsystem, showcasing energy flow and system management principles.

Current Progress

So far, our team has been in the brainstorming and component selection phase. We are carefully evaluating components that can work efficiently together within the limitations of size, power, and cost. This includes choosing buck converters (AP63203WU-7) for voltage regulation, Li-ion batteries for energy storage, solar panels for power input, and INA219/INA226 sensors for current and voltage monitoring. We are also in the process of designing the circuit schematics and transferring them into KiCad for PCB layout and simulation. This step allows us to visualize the interconnections between components and determine the optimal design approach for reliability and efficiency.

Future Plans

In the next stages, we plan to finalize the circuit design and proceed to PCB fabrication and testing. Once the power system is stable and verified, we aim to expand the project by integrating a reaction wheel system. This addition would allow us to simulate the attitude control of a real CubeSat, turning the project from a simple EPS demonstration into a complete, functional mock satellite model.

Power Consumption Analysis

Component	Voltage (V)	Current (mA)	Power (mW)	Quantity	Total Power (mW)	Notes
STM32 MCU	3.3	100+ mA	330	1	330	Main controller; active mode
INA219 / INA226	3.3	330µA	990		6.6	Monitoring sensors
AP63203WU-7 (Buck Converter) —		—	≈96% efficiency	2	—	5 V and 3.3 V rails
BQ24650 MPPT Charger	12	15µA	0.18	1	200	Solar charging & MPPT control
Li-Ion Battery (2S, 7.4V)	—	—	—	2	—	Energy storage
Reaction Wheel Motor	5	300–500	1500–2500	1	2000	Small DC motor w/ driver
Sensor Modules	3.3	10	33	2	66	Data acquisition sensors

Power Sources and Generation

Source	Nominal Voltage	Current (A)	Power (W)	Notes
Solar Panel (25 W)	12	2.08	25	Main power through MPPT
Battery Pack (2S Li-Ion, 2500 mAh)	7.4	—	18.5 Wh	Backup power and energy buffer

Estimated Battery Runtime

For two 3.7 V, 2500 mAh Li-ion batteries in series (≈18.5 Wh total): **Idle / Light Load (0.5 W):** ~37 hours **With Reaction Wheel (2.5 W):** ~7 hours **Continuous operation during sunlight:** Practically indefinite (solar-powered via MPPT).

Summary

The current EPS design efficiently supports the CubeSat prototype's electronics with low power draw. The 5 V and 3.3 V rails provide flexibility for digital and electromechanical loads. The solar + battery hybrid setup ensures continuous operation even in low light. The addition of the reaction wheel will increase power demand, highlighting the need for optimized energy management and battery capacity.

Future Work & Improvements

In the next phase, our team will focus on the following improvements and extensions: **1. PCB Fabrication & Testing:** Finalize the KiCad design and manufacture a functional PCB prototype for laboratory testing. **2. System Optimization:** Improve converter efficiency and implement dynamic power distribution using STM32 logic. **3. Telemetry & Data Logging:** Add communication modules to monitor system voltage, current, and temperature remotely. **4. Reaction Wheel Integration:** Design and test a reaction wheel subsystem to simulate CubeSat attitude control. **5. Final Model Assembly:** Combine all modules into a compact, educational CubeSat mock-up that visually demonstrates power flow and control.